Chapter 01: Introduction of Finance

1. What is Finance?

Finance is the study and management of money, investments, and other financial instruments. It includes all activities involving the flow of money:

1. Borrowing and Lending
2. Investing and Saving
3. Budgeting and Forecasting

Goal of Finance:

1. Maximize wealth for individuals or firms
2. Manage financial risks effectively

Main Areas:

- **Personal Finance:** Individual money management (e.g., budgeting, saving)
- **Corporate Finance:** Business financial decisions (e.g., investments, funding)
- **Public Finance:** Government revenue and expenditure (e.g., taxation, infrastructure funding)

2. Functions of Finance (with Examples):

1. **Investment Decision:** Choosing how to allocate funds to maximize returns.
   Example: A company invests in either expanding production or buying a new business.

2. **Financing Decision:** Choosing the best mix of equity (shares) and debt (loans) to fund operations.
   Example: Issuing shares vs. borrowing from banks to raise money.

3. **Dividend Decision:** How much profit to return to shareholders vs. reinvest in the business.
   Example: Paying 40% of profits as dividends and reinvesting 60%.

4. **Liquidity Management:** Ensuring there's enough cash to cover short-term needs.
   Example: Keeping a 6-month cash buffer or a credit line for emergencies.

5. **Risk Management:** Using strategies like hedging to protect against financial risks (price or currency fluctuations).
   Example: An airline locks in fuel prices to avoid future increases.

6. **Budgeting & Forecasting:** Planning income and expenses based on past and projected data.
   Example: A retail brand planning for increased sales during the holiday season.

7. **Cost Control:** Reducing unnecessary expenses to improve profits.
   Example: Using automation to reduce labor costs in manufacturing.

3. Why Wealth Maximization is Better than Profit Maximization

1. Long-term Growth Focus: Wealth maximization aims for sustainable, long-term growth rather than just quick short-term profits. It helps build lasting shareholder value over time.
Example:- Invests in R&D for future growth.

2. Risk and Time Management: It takes into account the risks and the timing of returns. A taka today is worth more than a taka tomorrow, and riskier ventures require careful evaluation.
Example: Avoids risky but quick-profit projects.

3. Focuses on Cash Flows: Wealth maximization is based on real cash flows rather than accounting profits, which ensures that the firm has actual liquidity to operate and grow.
Example: Limits credit sales to ensure cash flow.

4. Increases Market Value: It directly aims to increase the market value of shares, which reflects investor confidence and the overall financial health of the firm.
Example: Strong performance boosts stock price.

5. Balanced Stakeholder Interests: Unlike profit maximization, which focuses mainly on shareholders, wealth maximization considers employees, customers, and society to creating long-term trust and loyalty.
Example: Improves staff benefits for loyalty.

Examples:

1. A company consistently investing in innovation and employee welfare to grow over time that is Wealth Maximization
2. A firm cutting costs at the expense of product quality just to show short-term profits that is Profit Maximization, but risky in the long run.

4. Internal vs External Sources of Financing

Internal Financing (from within the business):

1. **Sources:**
    a. **Retained earnings**– Profits kept in the business instead of being distributed to shareholders.
    b. **Depreciation funds** – Money set aside from profits to replace worn-out assets.
    c. **Sale of unused assets** – Cash generated by selling non used equipment/property.
    d. **Working capital adjustments** – Freeing up cash by managing current assets and liabilities better.
    e.

2. **Advantages:**
    a. **No repayment or interest** – The business doesn't need to repay or pay interest on internal funds.
    b. **Full control over business decisions** – Owners retain full authority without external interference.
    c. **Faster and flexible access** – Funds can be used quickly without approval from outsiders.

3. **Disadvantages:**
    a. **Limited funds** – Internal sources usually provide a small amount of capital.
    b. **May slow expansion** – Growth might be delayed due to lack of sufficient funds.
    c. **Selling assets could harm operations** – Getting rid of useful assets might affect business efficiency.
4. ***Example:*** *Apple reinvesting profits into innovation.*

External Financing (sources outside business)

1. **External Financing Sources:**
    a. **Debt financing–** Borrowing money that must be repaid with interest.
    b. **Equity financing–** Raising funds by selling shares of the company.
    c. **Venture capital–** Investment from firms or individuals in exchange for ownership and high returns.
    d. **Government aid–** Non-repayable funds provided by the government or institutions.

2. **Advantages:**
    a. **Enables big investments** – External funding allows large-scale business projects.
    b. **Interest on debt may be tax-deductible** – Reduces taxable income, lowering tax burden.
    c. **No repayment in equity funding** – Investors share profits but don't require fixed repayments.

3. **Disadvantages:**
    a. **Debt increases financial risk** – Loans must be repaid even if profits are low.
    b. **Ownership dilution with equity** – Selling shares reduces control over the business.

c. **More regulations and formalities** – External funding involves legal and reporting requirements.

4. **Example:** Tesla issues new shares to fund expansion.

Risk And Return

ABC Capital is analyzing two stocks — Stock A and Stock B — to include in their investment portfolio. The firm wants to select the stock that offers the best return for the risk taken. They've gathered the annual returns (in %) for the last 5 years as shown below:

Year	Stock A (%)	Stock B (%)
1	11	9
2	13	12
3	7	10
4	15	14
5	9	8

Tasks:

a) Calculate the average (expected) return for both stocks.
b) Calculate the standard deviation (risk) of returns for both stocks.
c) Determine the return-to-risk ratio for each.
d) Based on your calculations, which stock should ABC Capital choose if they prefer a higher return per unit of risk?

📊 Risk and Return Analysis – XYZ Investments

XYZ Investments is considering adding one of two mutual funds — **Fund X** or **Fund Y** — to their portfolio. To make an informed decision, they've compiled the annual return data (in %) for the past **6 years** as shown below:

Year	Fund X (%)	Fund Y (%)
1	10	8
2	12	9
3	14	11
4	9	7
5	11	10
6	13	9

📝 Tasks:

a) Calculate the **average (expected) return** for both Fund X and Fund Y.

b) Calculate the **standard deviation (risk)** of returns for both funds.

c) Compute the **return-to-risk ratio** for each fund (Average Return ÷ Standard Deviation).

d) Based on your analysis, **which fund offers better performance per unit of risk**? Justify your answer.